

AWS Cloud Practical

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1. EC2 login using SSH and SSM

I. Launch an EC2 instance

II. Two login mechanisms: SSH and SSM

☰ [EC2](#) > [Instances](#) > Launch an instance ⓘ ⚙️ 🔍

Launch an instance ⓘ

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags ⓘ

Name

 [Add additional tags](#)

▼ **Application and OS Images (Amazon Machine Image) ⓘ**

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Recents

Quick Start

Amazon Linux



macOS



Ubuntu



Windows



Red Hat



SUSE Linux



De





[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

- Creating an EC2 instance named **ec2-iam-ssm-demo**, choosing an Amazon Linux AMI and creating a new key pair to enable SSH login.

Create key pair



Key pair name

Key pairs allow you to connect to your instance securely.

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type



RSA

RSA encrypted private and public key pair



ED25519

ED25519 encrypted private and public key pair

Private key file format



.pem

For use with OpenSSH



.ppk

For use with PuTTY



When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#)

Cancel

Create key pair

- Allow SSH port 22 from My IP.

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 122.171.225.37/32)

Remove

Type | [Info](#)

ssh

Protocol | [Info](#)

TCP

Port range | [Info](#)

22

Source type | [Info](#)

My IP

Name | [Info](#)

🔍 Add CIDR, prefix list or security gr

122.171.225.37/32 ✕

Description - *optional* | [Info](#)

e.g. SSH for admin desktop

Add security group rule

Instance summary for i-0113e23622b5aebfc (ec2-iam-ssm-demo) [Info](#)

[Connect](#) [Instance state](#) [Actions](#)

Updated less than a minute ago

Instance ID

i-0113e23622b5aebfc

IPv6 address

—

Hostname type

IP name: ip-172-31-15-126.eu-north-1.compute.internal

Answer private resource DNS name

—

Auto-assigned IP address

13.53.82.220 [Public IP]

IAM Role

—

Public IPv4 address

13.53.82.220 | [open address](#)

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-15-126.eu-north-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-04fbb5dfae5f66a87

Subnet ID

subnet-0c6e27b50c1eda (subnet1)

Private IPv4 addresses

172.31.15.126

Public DNS

ec2-13-53-82-220.eu-north-1.compute.amazonaws.com | [open address](#)

Elastic IP addresses

—

AWS Compute Optimizer finding

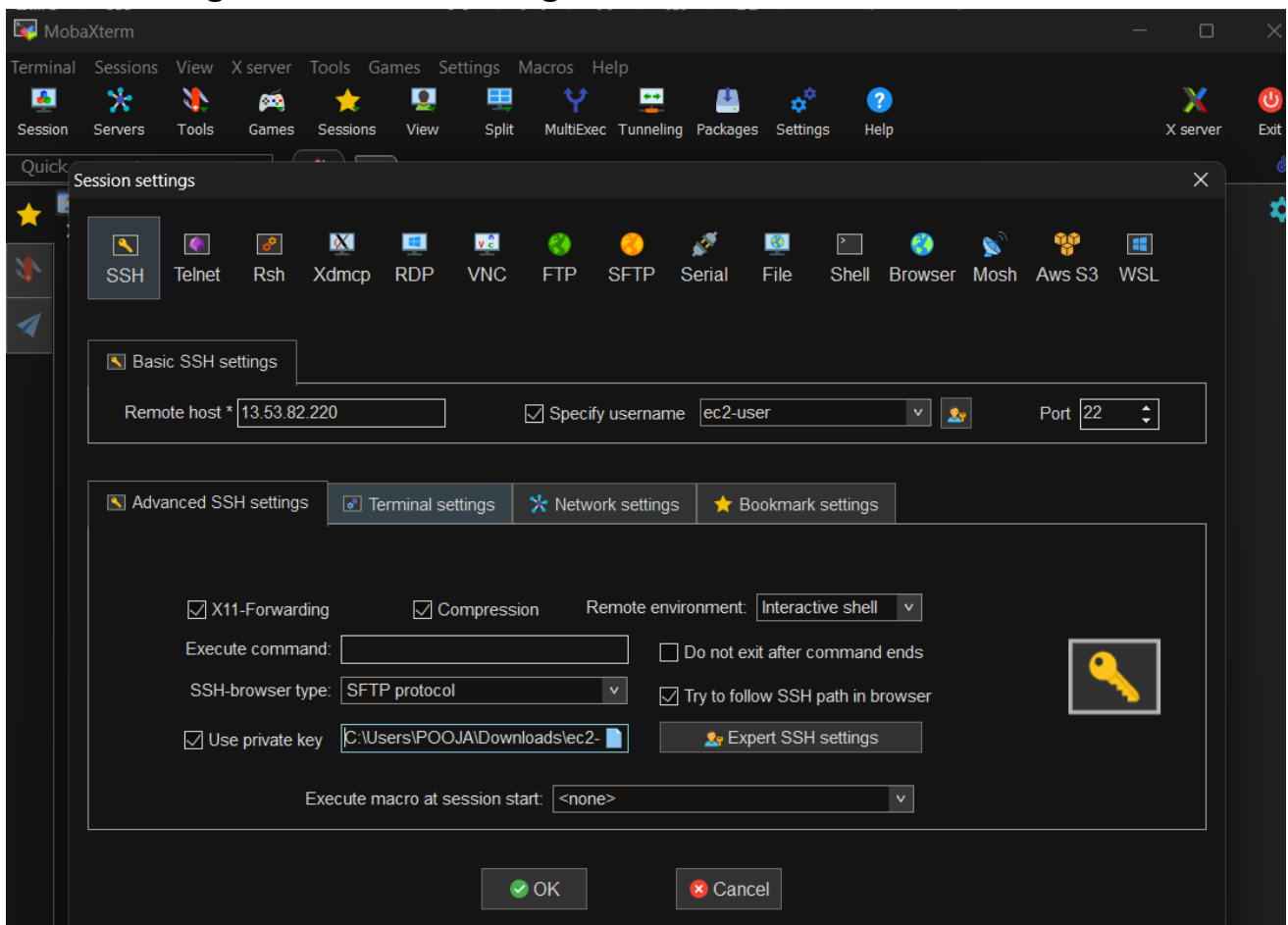
Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

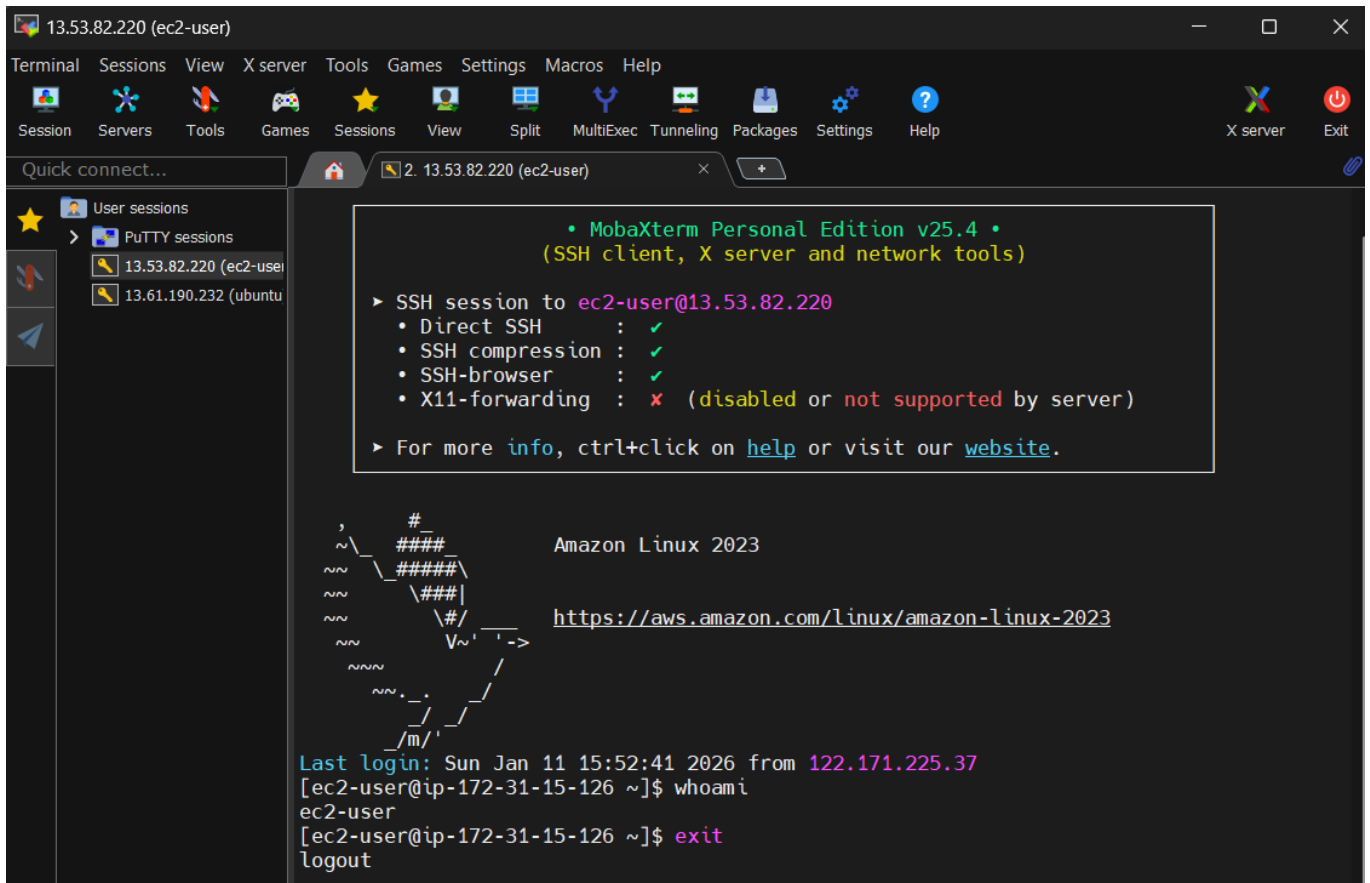
Auto Scaling Group name

—

i. EC2 login using SSH

- SSH login mechanism using MobaXterm.





ii. EC2 login using SSM

IAM > Roles > Create role

Step 1
● Select trusted entity
Step 2
○ Add permissions
Step 3
○ Name, review, and create

Select trusted entity [Info](#)

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
EC2

- For SSM mechanism initially we need to create an IAM role with permission **AmazonSSMManagedInstanceCore**, which will allow the EC2 instance to communicate with AWS Systems Manager.

☰ IAM > Roles > Create role ⓘ ⚙

Step 1
● Select trusted entity

Step 2
● **Add permissions**

Step 3
○ Name, review, and create

Add permissions ⓘ

Permissions policies (1/1110) ⓘ

Choose one or more policies to attach to your new role.

Filter by Type

🔍 AmazonSSMManaged ⓘ All types 1 match

☒	Policy name ⓘ	Type	Description
☒	 AmazonSSMManagedInstanceCore	AWS managed	The policy for A

▶ Set permissions boundary - optional

Cancel Previous Next

☰ IAM > Roles > Create role ⓘ ⚙

Step 2
● Add permissions

Step 3
● **Name, review, and create**

Role details

Role name
Enter a meaningful name to identify this role.

EC2-SSM-Role

Maximum 64 characters. Use alphanumeric and '+,.,@,-_' characters.

Description
Add a short explanation for this role.

Allows EC2 instances to call AWS services on your behalf.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+,. @-/\[\]!#\$%^&*()';:"'`

Step 1: Select trusted entities Edit

Trust policy

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sts:AssumeRole"
8       ],
9       "Principal": {
10        "Service": [
11          "ec2.amazonaws.com"
12        ]
13      }
14    ]
15  }

```

- Then attach Role to EC2 instance.

Instance summary for i-0113e23622b5aebfc (ec2-iam-ssm-demo) [Info](#)

[Connect](#)[Instance state ▼](#)[Actions ▲](#)

Updated less than a minute ago

Instance ID



i-0113e23622b5aebfc

Private IPv4 addresses



172.31.15.126

Instance state



Running

Hostname type

IP name: ip-172-31-15-126.eu-north-1.compute.internal

Instance diagnostics

Instance settings ▼

Networking ▼

Security ▲

Change security groups

Get Windows password

Modify IAM role

Image and templates ▼

Storage ▼

Monitor and troubleshoot ▼

compute.i

nternal

[EC2](#) > [Instances](#) > [i-0113e23622b5aebfc](#) > Modify IAM role



Modify IAM role [Info](#)

Attach an IAM role to your instance.

Instance ID



i-0113e23622b5aebfc (ec2-iam-ssm-demo)

IAM role

Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.

EC2-SSM-Role ▼



[Create new IAM role](#)

[Cancel](#)

[Update IAM role](#)

- Then try to connect to EC2 with Session Manager.

☰ [EC2](#) > [Instances](#) > [i-0113e23622b5aebfc](#) > Connect to instance

Connect [Info](#)

Connect to an instance using the browser-based client.

EC2 Instance Connect **Session Manager** SSH client EC2 serial console

Session Manager usage:

- Connect to your instance without SSH keys, a bastion host, or opening any inbound ports.
- Sessions are secured using an AWS Key Management Service key.
- You can log session commands and details in an Amazon S3 bucket or CloudWatch Logs log group.
- Configure sessions on the Session Manager [Preferences](#) [↗](#) page.

Cancel **Connect**

Session ID: root-k6flzerfhued6j88b432uvdjeu

 **Shortcuts**

Instance ID: i-0113e23622b5aebfc

Terminate

```
sh-5.2$ whoami
ssm-user
sh-5.2$
```

- I launched an EC2 instance, logged in using SSH with a private key, then configured an IAM role with **AmazonSSMManagedInstanceCore** and accessed the same EC2 using Session Manager without keys, demonstrating secure role-based access.